STOCKSENSE HARDWARE ARCHITECTURE

Component Shortlist & Technical Justifications — Target PCB Rev A (100x100mm)

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Target Controller: ESP32-DevKitC (38-Pin)

1. CORE MICROCONTROLLER (MCU) MODULE

SPECIFICATION PARAMETER	PRIMARY SELECTION: ESP32-DEVKITC (38-PIN)	ALTERNATIVE CONSIDERED: STM32F401 + ESP8266
Core Architecture	32-bit Dual-Core Xtensa LX6 @ 240 MHz	32-bit ARM Cortex-M4 @ 84 MHz
Memory	520 KB SRAM, 4 MB / 8 MB SPI Flash	96 KB SRAM, 512 KB Flash
Wireless Connectivity	Integrated Wi-Fi (802.11 b/g/n) & BLE v4.2	External ESP8266 via AT commands over UART
Power Distribution	Takes 5V on Pin 19; onboard LDO generates 3.3V (Pin 1)	Requires external 3.3V LDO regulator on PCB

Technical Rationale:

- **FreeRTOS Task Isolation:** Core 0 handles Wi-Fi stack & MQTTS (TLS 1.2 port 8883) keep-alives, while Core 1 handles real-time HX711 strain gauge sampling.
- **Onboard USB-UART & LDO:** Eliminates need for external USB programmer ICs and discrete 3.3V LDO regulators on the baseboard.

2. ANALOG FRONT-END (AFE) & WEIGHING SENSOR SUBSYSTEM

ROLE / SPECIFICATION	SELECTED COMPONENT	KEY OPERATING CHARACTERISTICS
ADC Converter	HX711 24-Bit Sigma-Delta ADC Breakouts (×3)	Selectable gain 64/128, 10Hz/80Hz sample rate, 2-wire serial protocol
Weight Transducer	Single-Point Wheatstone Bridge Strain Gauges	5 kg / 20 kg / 50 kg capacity options, 2.0 mV/V sensitivity
AFE Input Filtering	100 Ω Resistors (R11, R12) + 10 nF MLCC (C13..C15)	Low-pass RC differential noise suppression across INA+ / INA-

3. IDENTIFICATION & ENVIRONMENTAL SENSORS

SENSOR FUNCTION	SELECTED MODULE / PART	BUS INTERFACE & OPERATING RANGE
RFID Tag Reader	MFRC522 13.56 MHz RFID Reader Module	SPI Bus (up to 10 Mbit/s), ISO/IEC 14443A support (<50 ms scan time)
Climate Sensor	SHT30 / SHT31 Precision Temperature & Humidity	I2C Interface (Address 0x44), ±2% RH, ±0.2°C accuracy
Display / Auxiliary	0.96" OLED Display / BME280 Interface Socket	I2C Interface (Shared SDA/SCL bus with 4.7kΩ pull-ups R4, R5)

4. POWER MANAGEMENT SUBSYSTEM (PMIC & BATTERY BACKUP)

SUBSYSTEM BLOCK	SELECTED COMPONENT / MODULE	ROLE & OPERATING CHARACTERISTICS
Main DC Input	12V DC Barrel Jack (2.1mm) / 2-Pin Screw Terminal	Primary 12V DC wall adapter input rail
Step-Down Buck	Mini360 MP2307 Buck Converter Module	Steps down 12V DC rail to 5V DC (feeding ESP32 Pin 19)
Li-Ion Charger	TP5100 3S Buck Li-Ion Charger Board	Multi-cell charger configured for 3S (12.6V max charge voltage)
Backup Battery	3S 18650 Pack (11.1V Nominal, 2600 mAh)	Sustains full telemetry node power for ~28 hrs during grid outage
Battery Protection	3S 10A–20A BMS Board	Cell balancing, over-charge, over-discharge, and short-circuit protection
PowerPath OR-ing	SS34 Schottky Diodes (D_BAT1, D_MAINS1)	Dual-diode seamless power switching without MCU reboot or packet drop
Circuit Protection	AO3401A P-MOSFET + SMAJ15A TVS + 12V Zener	Reverse polarity supply protection & input ESD/overvoltage clamping

5. DISCRETE PASSIVES & BOARD HARDWARE

COMPONENT TYPE	VALUE / REFERENCE DESIGNATORS	PACKAGE & PLACEMENT FUNCTION
Resistors	10 kΩ (R2, R3, R6, R8, R10), 4.7 kΩ (R4, R5), 1 kΩ (R1)	0805 SMD — ESP32 EN pull-up, I2C pull-ups, transistor bias
Voltage Divider	30 kΩ (R_DIV1) & 10 kΩ (R_DIV2)	0805 SMD — High-precision mains voltage sensing into ADC pin
Capacitors	0.1 µF (C1,C4,C6,C9..C12), 4.7 µF (C3), 100 µF (C5)	0805 MLCC / 6.3mm Radial — Decoupling & bulk Wi-Fi burst buffering
Drivers & Indicators	BC847 NPN (Q3), Piezo Buzzer (BZ1), Status LEDs (D7, D8)	SOT-23 / 0805 — Audible alert driver and Power/Status LED indicators
Mechanicals	M3 Brass Standoffs (12mm) & Screw Mounting Holes	Four corner clearance mounts for 100x100mm shelf enclosure